

CMPE630: Digital Integrated Circuit Design

Laboratory Exercise 04

Introduction to Layout and Post-Extraction Simulation

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Lab Section 01
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By submitting this report, I attest that I neither have given nor have received any unauthorized assistance (including writing, collecting data, plotting figures, tables, or graphs, or using previous student reports as a reference), and I further acknowledge that giving or receiving such assistance will result in a failing grade for this course.

Your Signature: CGM

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1. Abstract

This laboratory exercise focused on the design and verification of a CMOS inverter layout using Cadence Virtuoso. The primary objectives were to create the inverter layout, perform a Design Rule Check (DRC) to ensure compliance with fabrication constraints, verify the Layout vs. Schematic (LVS) match, extract parasitic resistances and capacitances (RCX), and conduct post-layout simulations to analyze the impact of parasitic elements. The extracted layout was simulated and compared to the pre-layout simulation results from Lab 2, revealing the effects of **parasitic** on circuit performance. The post-extraction results indicated a rise time and a fall time **that show** a degradation compared to pre-layout simulations. The findings demonstrated that parasitic elements contributed to increased propagation delays and reduced maximum operating frequency. This emphasizes the importance of layout optimization in digital integrated circuit design to mitigate parasitic effects and improve overall circuit performance.

2. Design Methodology

2.1 CMOS Inverter Layout

The CMOS inverter layout was designed using Cadence Virtuoso, incorporating fundamental MOSFET design rules. The layout process began with the creation of a schematic diagram (**Error! Reference source not found.**), which provided a visual representation of the transistor connections before implementation. The stick diagram (**Error! Reference source not found.**) was translated into a physical design using standard layers, including oxide for defining the transistor gate, polysilicon for gate formation and routing, metal1 and metal2 for interconnections, and contacts and vias to establish connections between different layers.

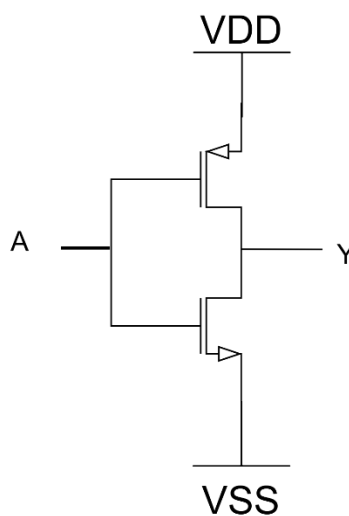


Figure 1: CMOS Inverter Schematics Diagram

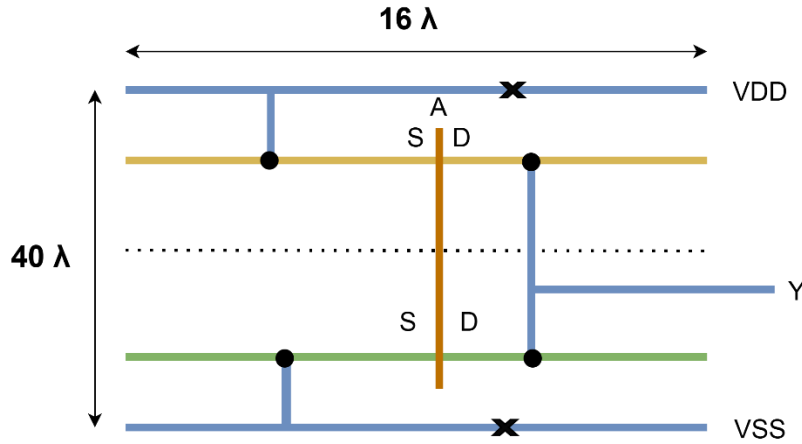


Figure 2: Stick Diagram of CMOS Inverter

The layout consists of multiple layers, each serving a specific function in defining the structure and electrical connectivity of the inverter. The active layer is composed of doped silicon, with n+ diffusion for NMOS and p+ diffusion for PMOS transistors. This layer forms the source and drain regions, enabling charge carrier flow. Above this, the gate oxide layer, made of silicon dioxide (SiO_2), acts as a dielectric insulator between the polysilicon gate and the underlying silicon channel, allowing for proper transistor operation. The polysilicon layer, used for gate formation, is made of doped polysilicon and serves as the control terminal for switching the transistor on and off. To establish electrical continuity between layers, contacts and vias, typically composed of tungsten or titanium, are used to connect diffusion regions, polysilicon, and metal layers.

For signal routing, the metal interconnect layers (Metal1 and Metal2), made of aluminum or copper, are used to connect the input (A), output (Y), and power rails (VDD and VSS). These metal layers provide low-resistance paths for electrical signals, ensuring efficient circuit operation. Finally, a passivation layer, made of silicon nitride or silicon dioxide, is applied to protect the circuit from contamination and environmental damage, enhancing long-term reliability. To optimize performance, area efficiency, and manufacturability, the PMOS transistor width was set to 240 nm, while the NMOS transistor width was 120 nm, ensuring balanced rise and fall times. The gate length was designed at 45 nm, following standard fabrication constraints. The estimated length from the stick diagram width and height from the stick diagram is 16λ and 40λ respectively. Thus, the area is estimated to be $64\lambda^2$. However, overall layout height and width were measured in Cadence Virtuoso to determine the total area occupied by the inverter, ensuring compliance with design rules and integration requirements. The width and height of the layout is 0.72nm and 1.9nm respectively, which gives an area of 1.368 nm^2 . Furthermore, I/O pins for input (A), output (Y), and power rails (VDD, VSS) were placed following industry layout standards. A place-and-route (P&R) boundary was defined for correct integration into automated design workflows. The final layout was carefully inspected to ensure compliance with required design rules before proceeding to verification checks.

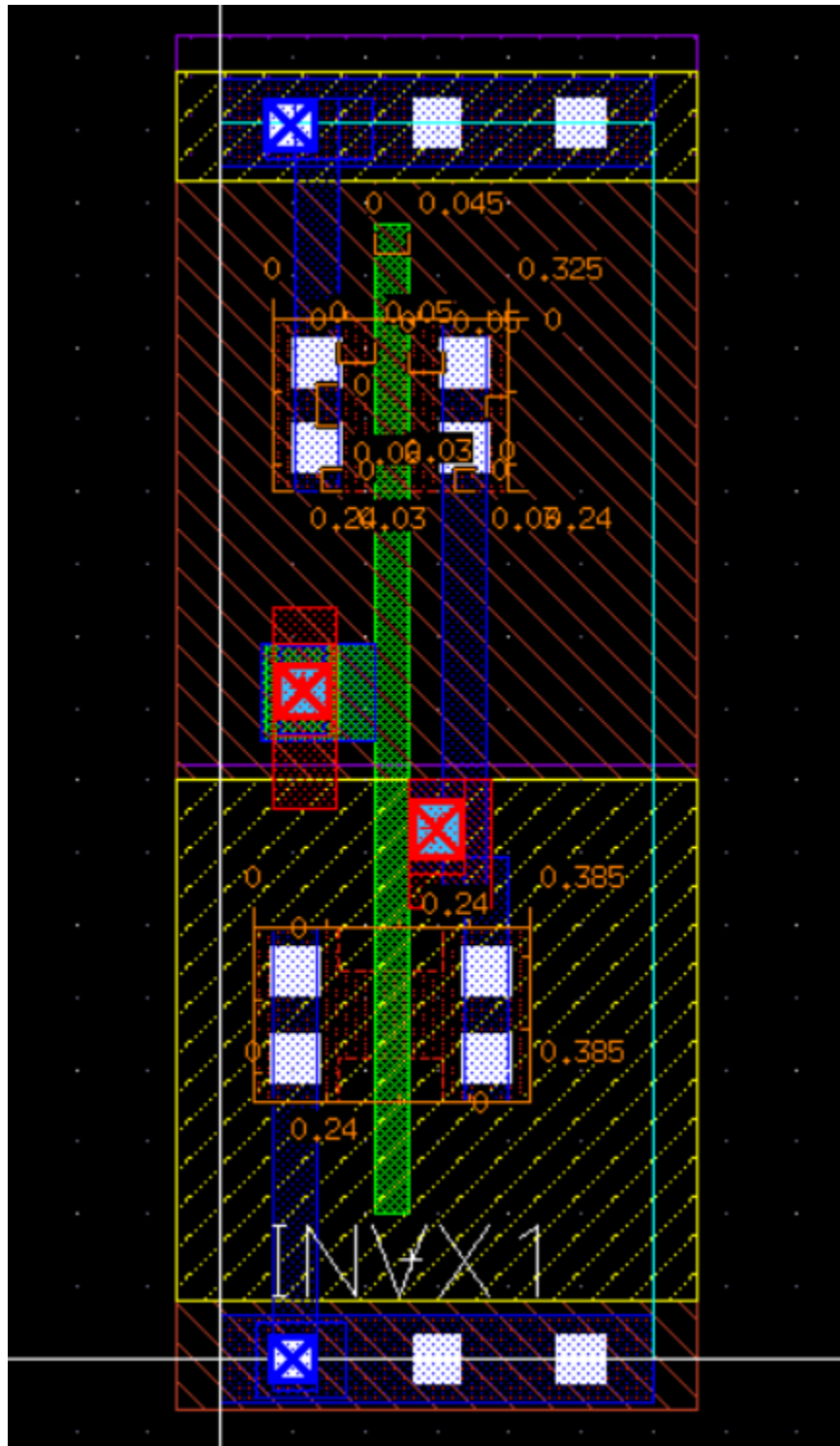


Figure 3: Final Inverter Layout

The Design Rule Check (DRC) was performed using the integrated Physical Verification System (PVS) in Cadence Virtuoso. DRC ensures that the layout conforms to the foundry's manufacturing constraints, preventing fabrication defects. Errors identified during the initial check included insufficient metal spacing, minimum width violations in polysilicon traces, and improper via placements. These errors were systematically corrected, and after several iterations, a clean DRC check was obtained, confirming compliance with design rules.

LVS was used to verify that the layout matched the original schematic design. This process involved extracting a netlist from the layout and comparing it with the schematic netlist. Discrepancies such as incorrect transistor connections and missing nodes were identified and resolved. A successful LVS check confirmed that the layout and schematic were functionally equivalent, ensuring that no unintended modifications were introduced during the layout design process.

Parasitic resistance and capacitance extraction was conducted using PVS-Quantus QRC. The extracted layout included parasitic resistances from metal interconnects and diffusion regions, as well as parasitic capacitances due to overlapping metal layers and junction effects. These parasitic elements were integrated into the extracted netlist for post-layout simulation to assess their impact on circuit performance.

3. Results and Analysis

3.1 Results of DRC, LVS and Extracted Layout

The results of the DRC verification, LVS and the picture of the extracted layout with parasitic elements is shown below in Figure 4, Figure 5 and **Error! Reference source not found.** respectively.

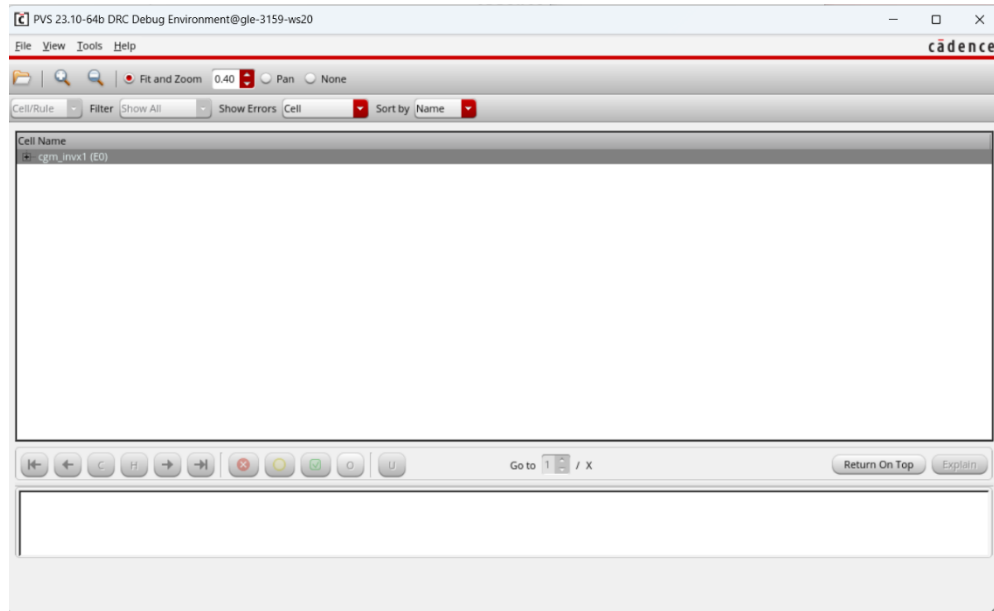
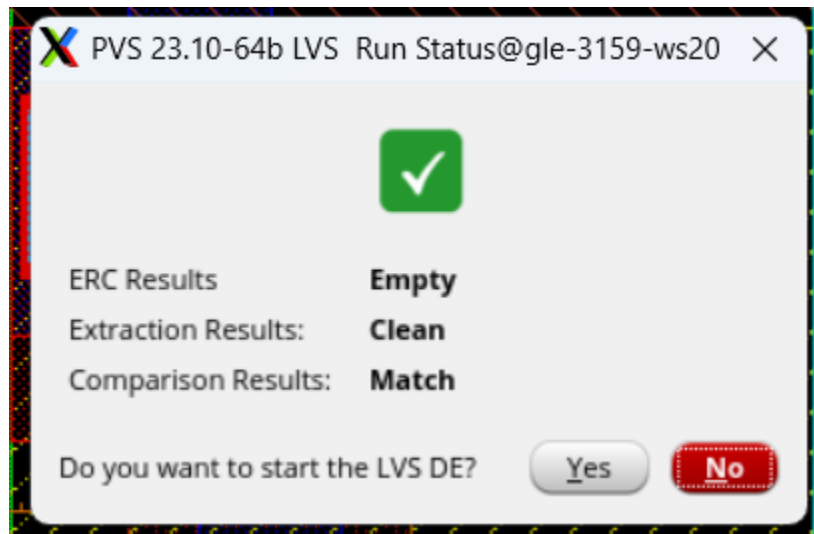


Figure 4: DRC Verification - No Errors Found



(a)

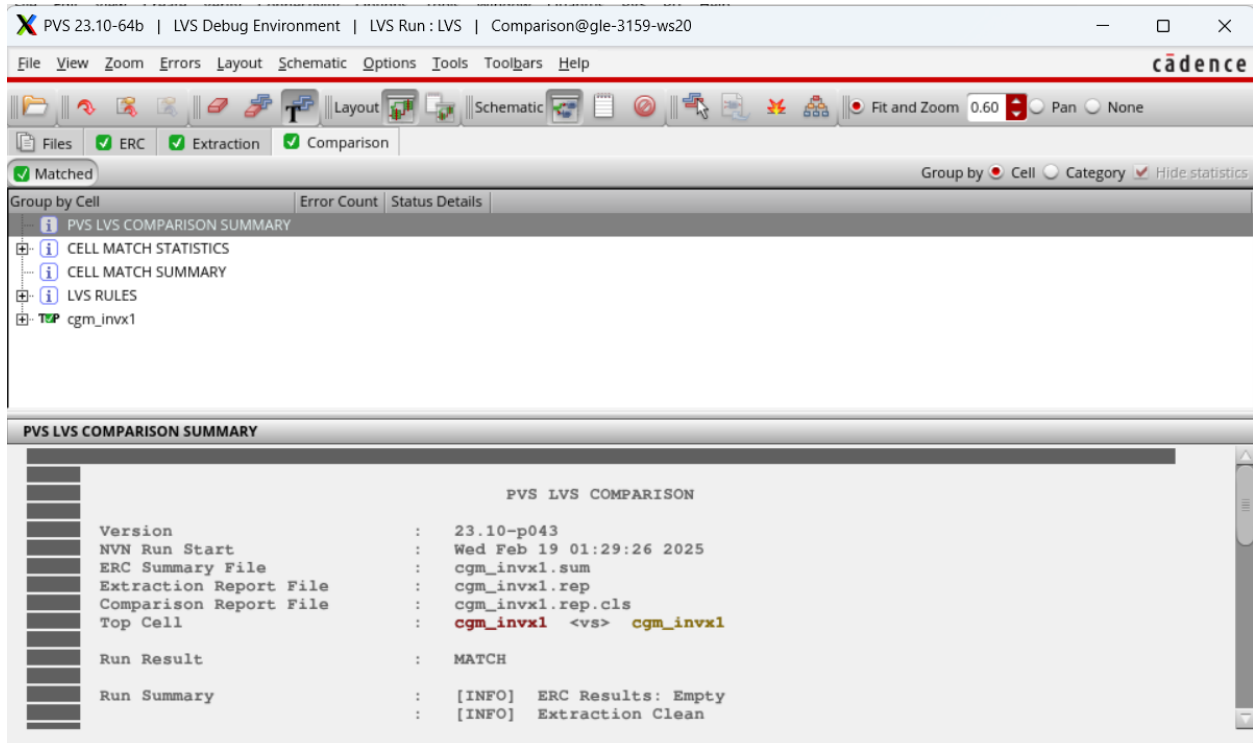


Figure 5: Extracted Layout with Parasitic Elements

The extracted layout of the CMOS inverter contains a total of one NMOS transistor and one PMOS transistor, consistent with the standard CMOS inverter design.

In addition to the transistors, the extracted layout includes parasitic resistors and capacitors that arise from physical fabrication constraints. The layout extraction process revealed the presence of 10 parasitic resistors and 12 parasitic capacitors, which are introduced due to interconnect resistance, diffusion resistance, and overlapping conductive layers. A detailed table listing all the extracted components from the CMOS inverter layout, including transistors, parasitic resistors, and parasitic capacitors, along with their respective values and connections is shown in Table 1(a), 1(b) and 1(c).

Table 1(a) Transistors

Component	Type	Width (W)	Length (L)	Connections
PM0	PMOS	240 nm	45 nm	Drain (Y), Gate (A), Source (VDD)
NM0	NMOS	120 nm	45 nm	Drain (Y), Gate (A), Source (VSS)

Table 1 (b) Parasitic Resistors

Component	Resistance (Ω)	Connections (Between Nodes)
Rj1	0.5023	A $\rightarrow$ \6:A

Rj2	0.5	$Y \rightarrow \backslash 2:Y$
Rk1	45	$\backslash 3:A \rightarrow \backslash 6:A$
Rk5	31.2754	$VDD \rightarrow \backslash 2:VDD$
Rk7	37.8171	$VSS \rightarrow \backslash 2:VSS$
Rk2	31.4078	$\backslash 1:Y \rightarrow \backslash 2:Y$
Rk3	37.6581	$\backslash 2:Y \rightarrow \backslash 3:Y$
RI1	101.157	$\backslash 1:A \rightarrow \backslash 2:A$
RI2	8.2281	$\backslash 2:A \rightarrow \backslash 3:A$
RI3	143.464	$\backslash 2:A \rightarrow \backslash 4:A$

Table 1 © Parasitic Capacitors

Component	Capacitance (F)	Connections (Between Nodes)
C1	3.08221e-17	$A \rightarrow VSS$
C2	8.66317e-17	$VDD \rightarrow VSS$
C3	2.32118e-17	$Y \rightarrow VSS$
C4	3.64039e-17	$\backslash 1:A \rightarrow VSS$
C5	3.89858e-17	$\backslash 4:A \rightarrow VSS$
C6	1.07027e-17	$\backslash 3:A \rightarrow VSS$
C7	6.14295e-17	$\backslash 2:Y \rightarrow VSS$
C8	4.13967e-17	$\backslash 6:A \rightarrow VSS$
C9	4.9106e-17	$\backslash 1:Y \rightarrow VSS$
C10	3.23836e-17	$\backslash 2:VDD \rightarrow VSS$
C11	3.40284e-17	$\backslash 3:Y \rightarrow VSS$
C12	2.65702e-17	$\backslash 2:A \rightarrow VSS$

The presence of 10 parasitic resistors and 12 parasitic capacitors in the extracted layout indicates interconnect resistance, junction capacitances, and gate overlap capacitances, which contribute to increased propagation delay. These parasitics must be considered in post-layout simulations to ensure the inverter meets timing constraints.

3.2 Simulation Comparisons

The inverter was tested under the same conditions as in Lab 2, with nominal conditions at $VDD = 1.1V$ and $Temp = 25^{\circ}C$, worst-case conditions at $VDD = 1.0V$ and $Temp = 125^{\circ}C$, and load capacities of 0 fF (unloaded) and 120 fF (loaded). The extracted parasitics impacted inverter performance, increasing rise time, fall time, and propagation delays. The transient simulation waveform from Lab2 is shown below in Figure 6.

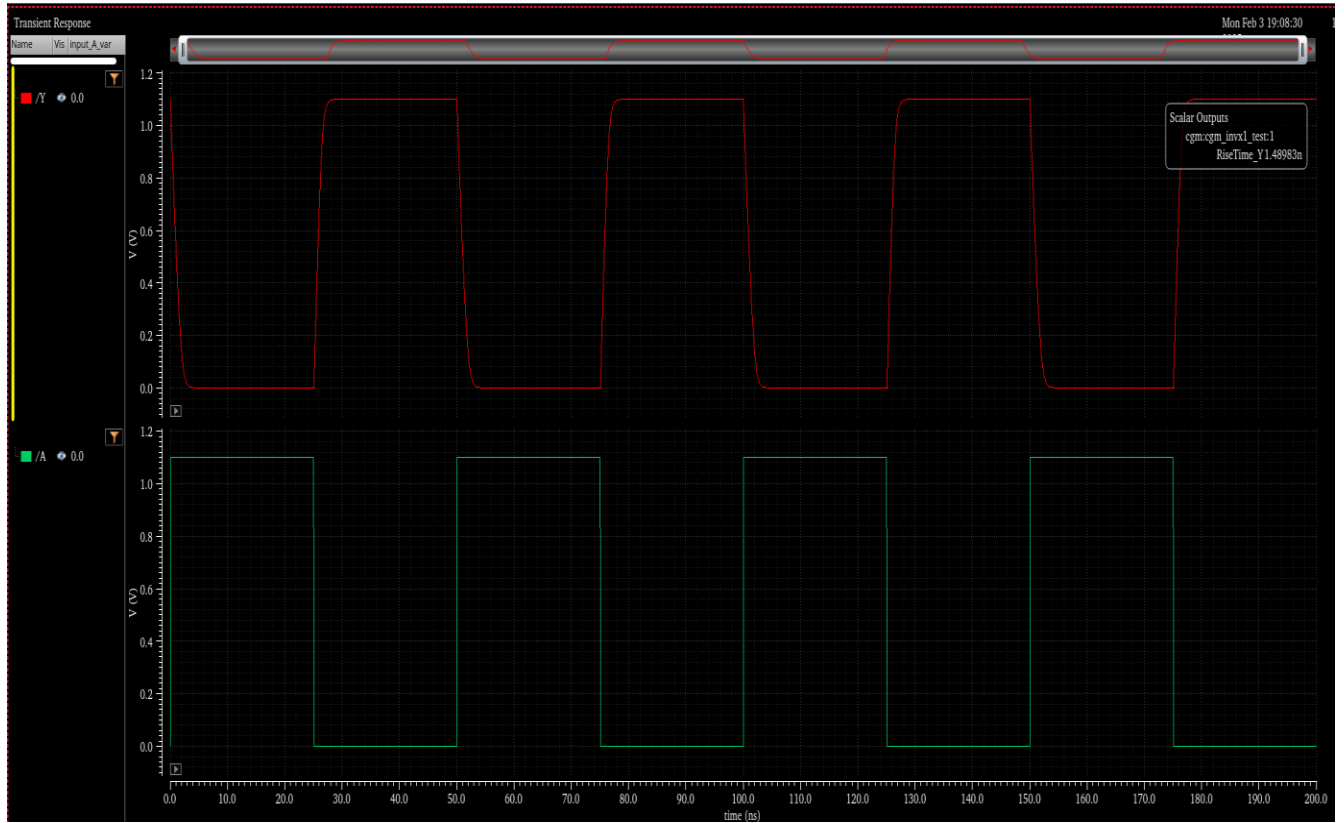


Figure 6: CMOS Inverter Transient Simulation Waveform

The Pre-extraction test results (same as Lab 2) in the ADE XL editor is shown below in figure

Test	Output	Plot All	Nominal	Spec	Weight	Pass/Fail
Nominal_RiseTime	/A					
Nominal_RiseTime	Nominal_RiseTime		10.18p			
Nominal_RiseTime	Nominal_RiseTime_Delay		11p			
Nominal_RiseTime_Load	/Y					
Nominal_RiseTime_Load	/A					
Nominal_RiseTime_Load	Nominal_RiseTime_Load		1.783n			
Nominal_RiseTime_Load	Nominal_RiseTime_Load_Delay		882.5p			
Nominal_FallTime	/Y					
Nominal_FallTime	/A					
Nominal_FallTime	Nominal_Falltime		10.81p			
Nominal_FallTime	Nominal_Falltime_Delay		13.6p			
Nominal_FallTime_Load	/Y					
Nominal_FallTime_Load	/A					
Nominal_FallTime_Load	Nominal_Falltime_Load		2.283n			
Nominal_FallTime_Load	Nominal_Falltime_Load_Delay		1.149n			
Worst_Case_Risetime	/Y					
Worst_Case_Risetime	/A					
Worst_Case_Risetime	Worst_Case_Risetime		12.78p			
Worst_Case_Risetime	Worst_Case_Risetime_Delay		15.4p			
Worst_Case_Risetime_Load	/Y					
Worst_Case_Risetime_Load	/A					
Worst_Case_Risetime_Load	Worst_Case_Risetime_Load		2.757n			
Worst_Case_Risetime_Load	Worst_Case_Risetime_Load_Delay		1.364n			
Worst_Case_Falltime	/Y					
Worst_Case_Falltime	/A					
Worst_Case_Falltime	Worst_Case_Falltime		13.38p			
Worst_Case_Falltime	Worst_Case_Falltime_Delay		18.92p			
Worst_Case_Falltime_Load	/Y					
Worst_Case_Falltime_Load	/A					
Worst_Case_Falltime_Load	Worst_Case_Falltime_Load		3.864n			
Worst_Case_Falltime_Load	Worst_Case_Falltime_Load_Delay		1.938n			

Figure 7: Pre-extraction test results

The post extraction test result is shown below in Figure 8.

Test	Output	Nominal	Spec	Weight	Pass/Fail
Nominal_RiseTime	/A				
Nominal_RiseTime	Nominal_RiseTi...	13.76p			
Nominal_RiseTime	Nominal_RiseTi...	14.37p			
Nominal_RiseTime_Load	/Y				
Nominal_RiseTime_Load	/A				
Nominal_RiseTime_Load	Nominal_RiseTi...	1.805n			
Nominal_RiseTime_Load	Nominal_RiseTi...	891.8p			
Nominal_FallTime	/Y				
Nominal_FallTime	/A				
Nominal_FallTime	Nominal_Falltime	14.13p			
Nominal_FallTime	Nominal_Falltim...	17.06p			
Nominal_FallTime_Load	/Y				
Nominal_FallTime_Load	/A				
Nominal_FallTime_Load	Nominal_Falltim...	2.256n			
Nominal_FallTime_Load	Nominal_Falltim...	1.134n			
Worst_Case_Risetime	/Y				
Worst_Case_Risetime	/A				
Worst_Case_Risetime	Worst_Case_Ris...	16.66p			
Worst_Case_Risetime	Worst_Case_Ris...	19.38p			
Worst_Case_Risetime_Load	/Y				
Worst_Case_Risetime_Load	/A				
Worst_Case_Risetime_Load	Worst_Case_Ris...	2.797n			
Worst_Case_Risetime_Load	Worst_Case_Ris...	1.381n			
Worst_Case_Falltime	/Y				
Worst_Case_Falltime	/A				
Worst_Case_Falltime	Worst_Case_Fall...	17.99p			
Worst_Case_Falltime	Worst_Case_Fall...	22.98p			
Worst_Case_Falltime_Load	/Y				
Worst_Case_Falltime_Load	/A				
Worst_Case_Falltime_Load	Worst_Case_Fall...	3.877n			
Worst_Case_Falltime_Load	Worst_Case_Fall...	1.926n			

Figure 8: Post extraction results

The post extraction data is summarized below in

Table 2.

Table 2: Inverter Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime (s)	FallTime (s)	T _{PHL} (s)	T _{PLH} (s)
Nominal	1.1	25	0	13.76p	14.13p	17.06p	14.37p
	1.1	25	120	1.805 n	2.256n	1.134n	891.8p
Worst Case	1	125	0	16.66p	17.99p	22.98p	19.38p
	1	125	120	2.797n	3.877n	1.926n	1.381n

The timing simulations of the extracted netlist show that the post-extraction simulation is slower compared to the results from Lab 2 (pre-extraction). This is evident from the increase in rise time, fall time, and propagation delays (T_{PHL} and T_{PLH}) across all test conditions. The slower timing in the post-extraction simulation is due to the introduction of parasitic resistances and capacitances that were not present in the pre-extraction results. These parasitics originate from interconnect resistance, diffusion resistance, and unwanted capacitances between metal layers and transistor terminals. The presence of these additional elements increases the RC delay, requiring more time for signals to transition between logic states. Additionally, higher temperatures (125°C) further contribute to increased delay by reducing carrier mobility, which increases resistance and slows switching speed.

The maximum input frequency, $F_{input,max}$ represents the highest frequency at which the inverter can reliably process input transitions. It is given by the equation:

$$F_{input,max} = 1/(t_{rise} + t_{fall})$$

Similarly, the maximum throughput frequency, $F_{throughput,max}$ is determined by the total propagation delay and is given by:

$$F_{throughput,max} = 1/(T_{P,HL} + T_{P,LH})$$

Using the extracted timing values from both nominal and worst-case conditions, the maximum operating frequencies were computed. The results are summarized in the Table 3 below.

Table 3: Maximum Operating Frequency Calculations

	VDD (V)	Temp (C)	Load (fF)	$F_{input,max}$ (Hz)	$F_{throughput,max}$ (Hz)
Inverter	1	125	0	2.87×10^{10}	2.36×10^{10}
	1	125	120	1.51×10^8	3.02×10^8

The calculations for the percentage decrease in maximum operating frequency and throughput are as follows:

$$Percentage\ decrease = ((F_{unloaded} - F_{loaded}) / F_{unloaded}) \times 100$$

1. Maximum Input Frequency ($F_{input,max}$)

Unloaded: 2.87×10^{10} Hz

Loaded (120 fF): 1.51×10^8 Hz

$$\begin{aligned}
 Percentage\ Decrease &= \left(\frac{(2.87 \times 10^{10} - 1.51 \times 10^8)}{2.87 \times 10^{10}} \right) \times 100 \\
 &= \frac{(2.87 \times 10^{10} - 0.151 \times 10^{10})}{2.87 \times 10^{10}} \times 100 \\
 &= 99.47\%
 \end{aligned}$$

2. Maximum Throughput Frequency ($F_{throughput,max}$)

Unloaded: 2.36×10^{10} Hz

Loaded (120 fF): 3.02×10^8 Hz

$$\begin{aligned}
 Percentage\ Decrease &= \left(\frac{(2.36 \times 10^{10} - 3.02 \times 10^8)}{2.36 \times 10^{10}} \right) \times 100 \\
 &= \frac{(2.36 \times 10^{10} - 0.302 \times 10^{10})}{2.36 \times 10^{10}} \times 100 \\
 &= 98.72\%
 \end{aligned}$$

These results highlight the significant impact of capacitive loading and worst-case conditions on inverter performance, emphasizing the importance of minimizing parasitic capacitances in high-speed circuit design.

The calculation of the percentage speed change (post-extraction vs. pre-extraction) is shown below:

Using the Formula:

$$\text{Percentage Change} = ((\text{Pre-extraction} - \text{Post-extraction}) / \text{Pre-extraction}) \times 100$$

1. Rise Time

$$\begin{aligned} - \text{Nominal (0 fF load)} &= ((10.08p - 13.76p) / 10.08p) \times 100 \\ &= (-3.68p / 10.08p) \times 100 \\ &= -36.5\% (\text{slower}) \end{aligned}$$

$$\begin{aligned} - \text{Nominal (120 fF load)} &= ((1.783n - 1.805n) / 1.783n) \times 100 \\ &= (-0.022n / 1.783n) \times 100 \\ &= -1.23\% (\text{slower}) \end{aligned}$$

$$\begin{aligned} - \text{Worst Case (0 fF load)} &= ((12.78p - 16.66p) / 12.78p) \times 100 \\ &= (-3.88p / 12.78p) \times 100 \\ &= -30.4\% (\text{slower}) \end{aligned}$$

$$\begin{aligned} - \text{Worst Case (120 fF load)} &= ((2.757n - 2.797n) / 2.757n) \times 100 \\ &= (-0.04n / 2.757n) \times 100 \\ &= -1.45\% (\text{slower}) \end{aligned}$$

2. Fall Time

$$\begin{aligned} - \text{Nominal (0 fF load)} &= ((10.81p - 14.13p) / 10.81p) \times 100 \\ &= (-3.32p / 10.81p) \times 100 \\ &= -30.7\% (\text{slower}) \end{aligned}$$

$$\begin{aligned} - \text{Nominal (120 fF load)} &= ((2.283n - 2.256n) / 2.283n) \times 100 \\ &= (0.027n / 2.283n) \times 100 \\ &= +1.18\% (\text{faster}) \end{aligned}$$

$$\begin{aligned} - \text{Worst Case (0 fF load)} &= ((13.38p - 17.99p) / 13.38p) \times 100 \\ &= (-4.61p / 13.38p) \times 100 \\ &= -34.5\% (\text{slower}) \end{aligned}$$

$$\begin{aligned} - \text{Worst Case (120 fF load)} &= ((3.864n - 3.877n) / 3.864n) \times 100 \\ &= (-0.013n / 3.864n) \times 100 \\ &= -0.34\% (\text{slower}) \end{aligned}$$

3. Propagation Delay (T_{PHL})

$$\begin{aligned} \text{- Nominal (0 fF load)} &= ((13.6p - 17.06p) / 13.6p) \times 100 \\ &= (-3.46p / 13.6p) \times 100 \\ &= -25.4\% \text{ (slower)} \end{aligned}$$

$$\begin{aligned} \text{- Nominal (120 fF load)} &= ((1.149n - 1.134n) / 1.149n) \times 100 \\ &= (0.015n / 1.149n) \times 100 \\ &= +1.3\% \text{ (faster)} \end{aligned}$$

$$\begin{aligned} \text{- Worst Case (0 fF load)} &= ((18.92p - 22.98p) / 18.92p) \times 100 \\ &= (-4.06p / 18.92p) \times 100 \\ &= -21.4\% \text{ (slower)} \end{aligned}$$

- Worst Case (120 fF load):

$$\begin{aligned} &((1.938n - 1.926n) / 1.938n) \times 100 \\ &= (0.012n / 1.938n) \times 100 \\ &= +0.62\% \text{ (faster)} \end{aligned}$$

4. Propagation Delay (T_{PLH})

- Nominal (0 fF load):

$$\begin{aligned} &= ((11p - 14.37p) / 11p) \times 100 \\ &= (-3.37p / 11p) \times 100 \\ &= -30.6\% \text{ (slower)} \end{aligned}$$

$$\begin{aligned} \text{- Nominal (120 fF load)} &= ((882.5p - 891.8p) / 882.5p) \times 100 \\ &= (-9.3p / 882.5p) \times 100 \\ &= -1.06\% \text{ (slower)} \end{aligned}$$

$$\begin{aligned} \text{- Worst Case (0 fF load)} &= ((15.4p - 19.38p) / 15.4p) \times 100 \\ &= (-3.98p / 15.4p) \times 100 \\ &= -25.9\% \text{ (slower)} \end{aligned}$$

$$\begin{aligned} \text{- Worst Case (120 fF load)} &= ((1.364n - 1.381n) / 1.364n) \times 100 \\ &= (-0.017n / 1.364n) \times 100 \\ &= -1.25\% \text{ (slower)} \end{aligned}$$

The calculation of speedup/slowdown calculation (post-extraction vs. pre-extraction) is as follows:

Using the formula:

$$\text{Speed Change (\%)} = ((\text{Post-extraction} - \text{Pre-extraction}) / \text{Pre-extraction}) \times 100$$

1. Maximum Input Frequency (F_{input,max})

- Pre-Extraction: 1.50×10^8 Hz

- Post-Extraction: 1.51×10^8 Hz

$$\begin{aligned}
- \text{Speed Change} &= ((1.51 \times 10^8 - 1.50 \times 10^8) / 1.50 \times 10^8) \times 100 \\
&= (0.01 \times 10^8 / 1.50 \times 10^8) \times 100 \\
&= +0.67\% \text{ (slight speedup)}
\end{aligned}$$

2. Maximum Throughput Frequency ($F_{\text{throughput,max}}$)

- Pre-Extraction: 3.03×10^8 Hz

- Post-Extraction: 3.02×10^8 Hz

$$\begin{aligned}
- \text{Speed Change} &: ((3.02 \times 10^8 - 3.03 \times 10^8) / 3.03 \times 10^8) \times 100 \\
&= (-0.01 \times 10^8 / 3.03 \times 10^8) \times 100 \\
&= -0.33\% \text{ (slight slowdown)}
\end{aligned}$$

4. Conclusion

The post-layout simulation results demonstrated the impact of parasitic elements on inverter performance. The extracted inverter exhibited increased propagation delay, reduced maximum operating frequency, and degraded switching characteristics compared to pre-layout simulations. These findings highlight the importance of layout optimization in mitigating parasitic effects and ensuring robust digital circuit performance. Future improvements could include reducing parasitic capacitances by optimizing metal routing and minimizing diffusion areas.

5. Questions and Answers

What do the extracted resistors and capacitors represent?

The extracted resistances correspond to poly, metal, and diffusion regions, while capacitances arise from inter-layer and junction parasitics.

Draw the equivalent circuit of the extracted view with parasitics.

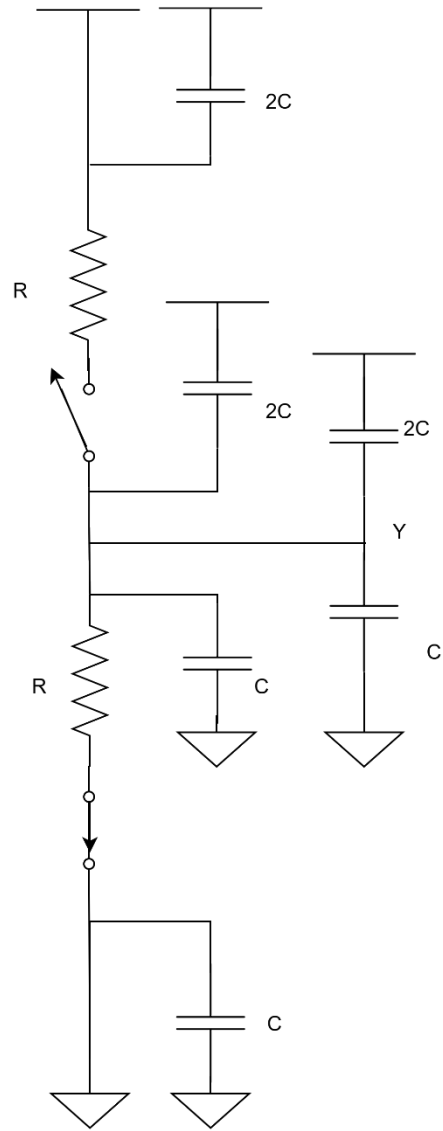


Figure 6: Equivalent Circuit with Extracted Parasitics

6. Signoff Sheet

8 Signoff Sheet

Table 3: Inverter Layout

Students Name:

Gloria Mbaka

Section: 1

Prelab	Point Value	Points Earned	Date
Inverter Stick Diagram	10	10	JE 2/19
Demo	Point Value	Points Earned	Date
DRC check	25	25	JE 2/19
LVS check	25	25	↓
Parasitic RC RCX	20	20	
Post-Extraction Results	20	20	
Total Points	100		